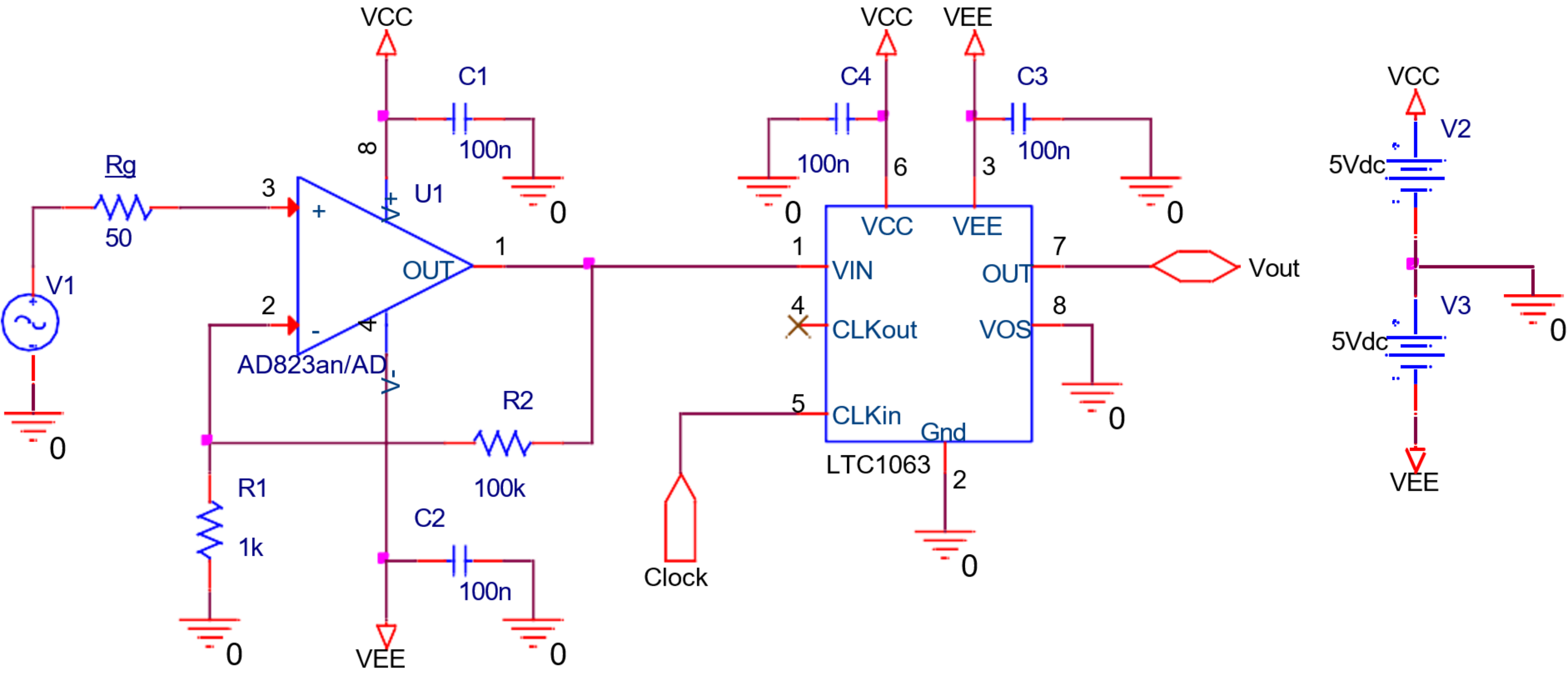
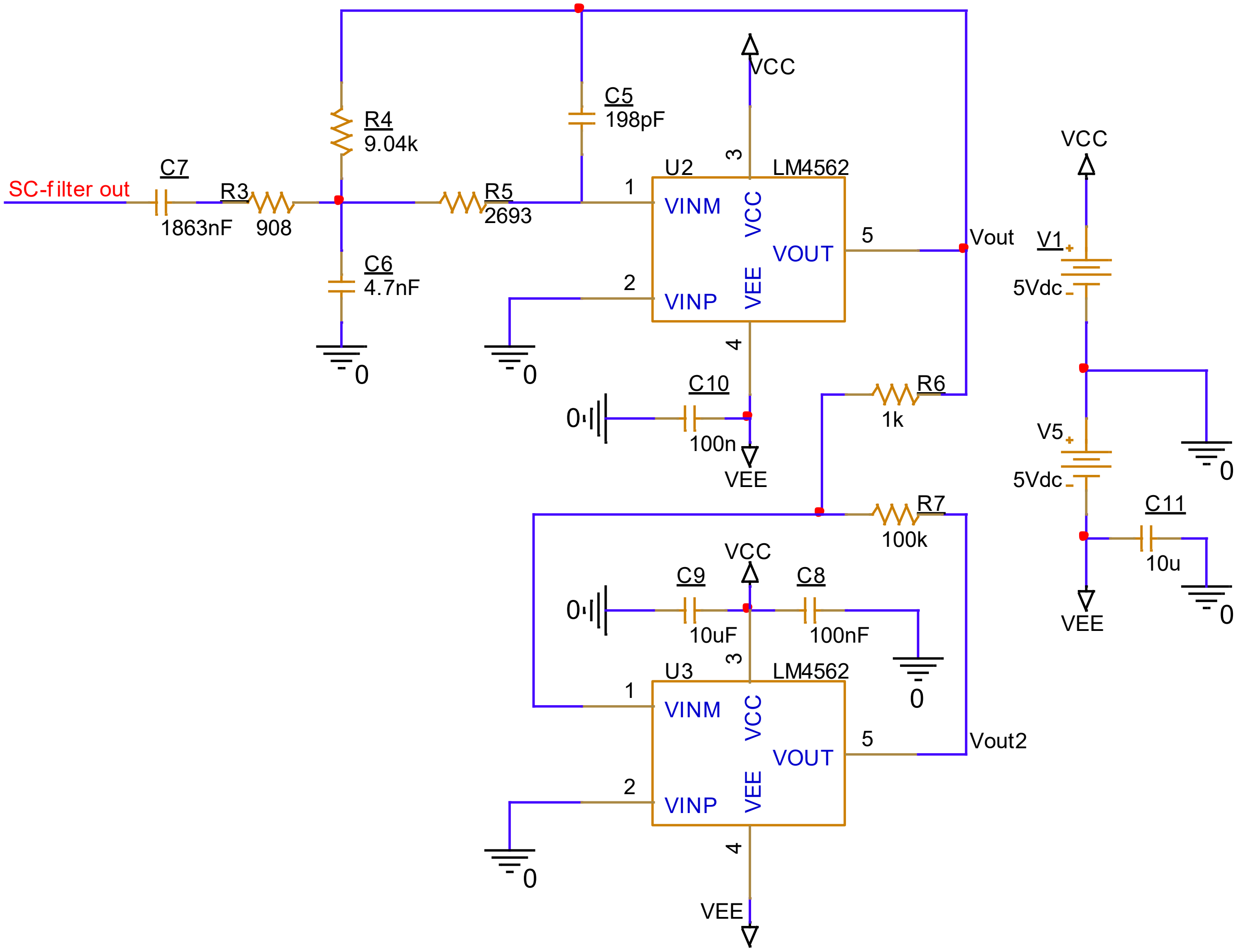
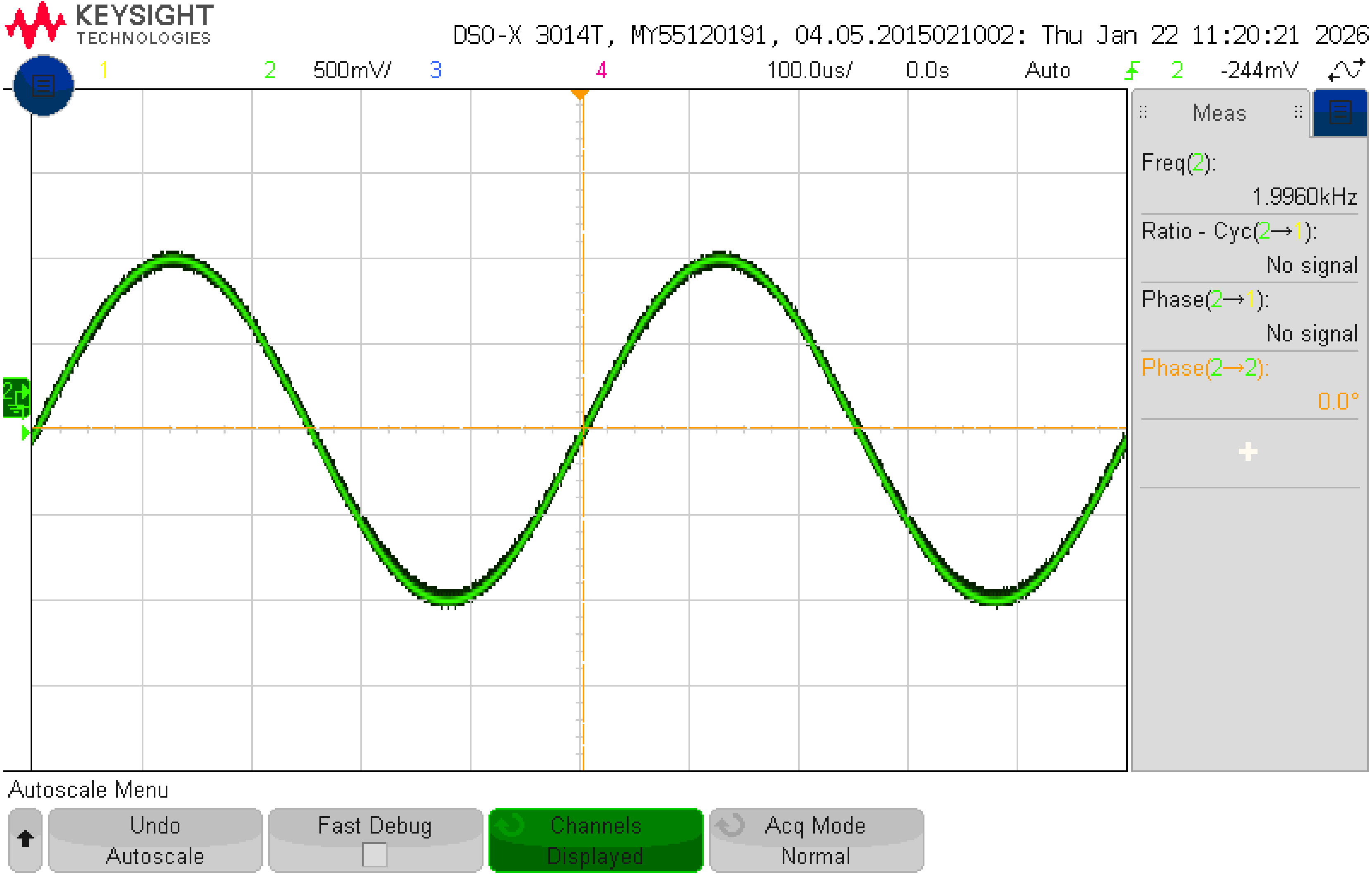
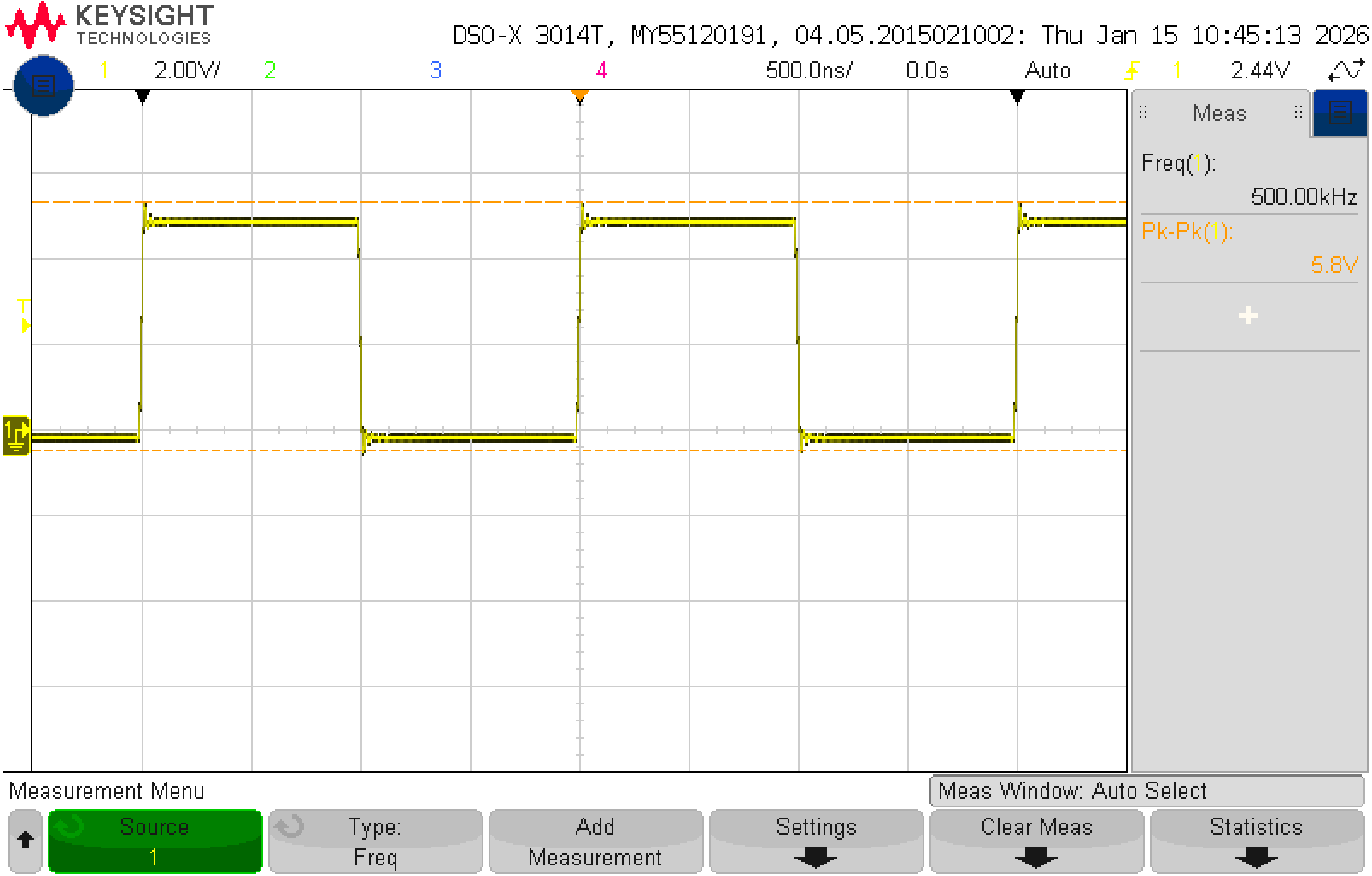


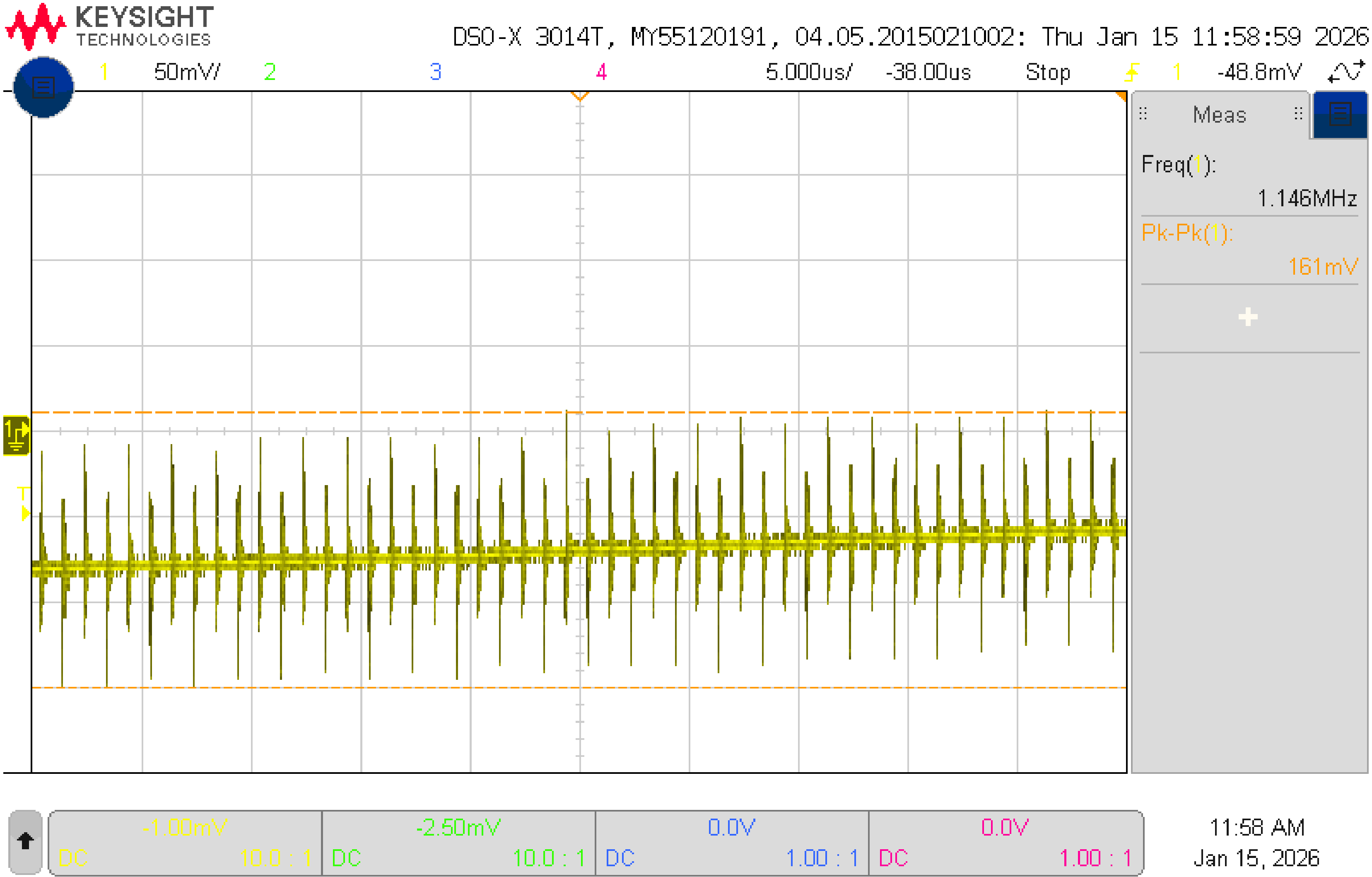
VCC
C1
100n
Rg
50
3
8
U1
V+
0
+
1
OUT
V1
2
-
4
AD823an/AD
V-
R2
100k
R1
1k
C2
100n
0
VEE
0
0
VCC
VEE
C4
C3
100n
100n
6
3
0
0
VCC
VEE
1
VIN
7
OUT
Vout
4
CLKout
8
VOS
5
CLKin
Gnd
0
LTC1063
2
Clock
0
VCC
V2
5Vdc
V3
5Vdc
0
VEE

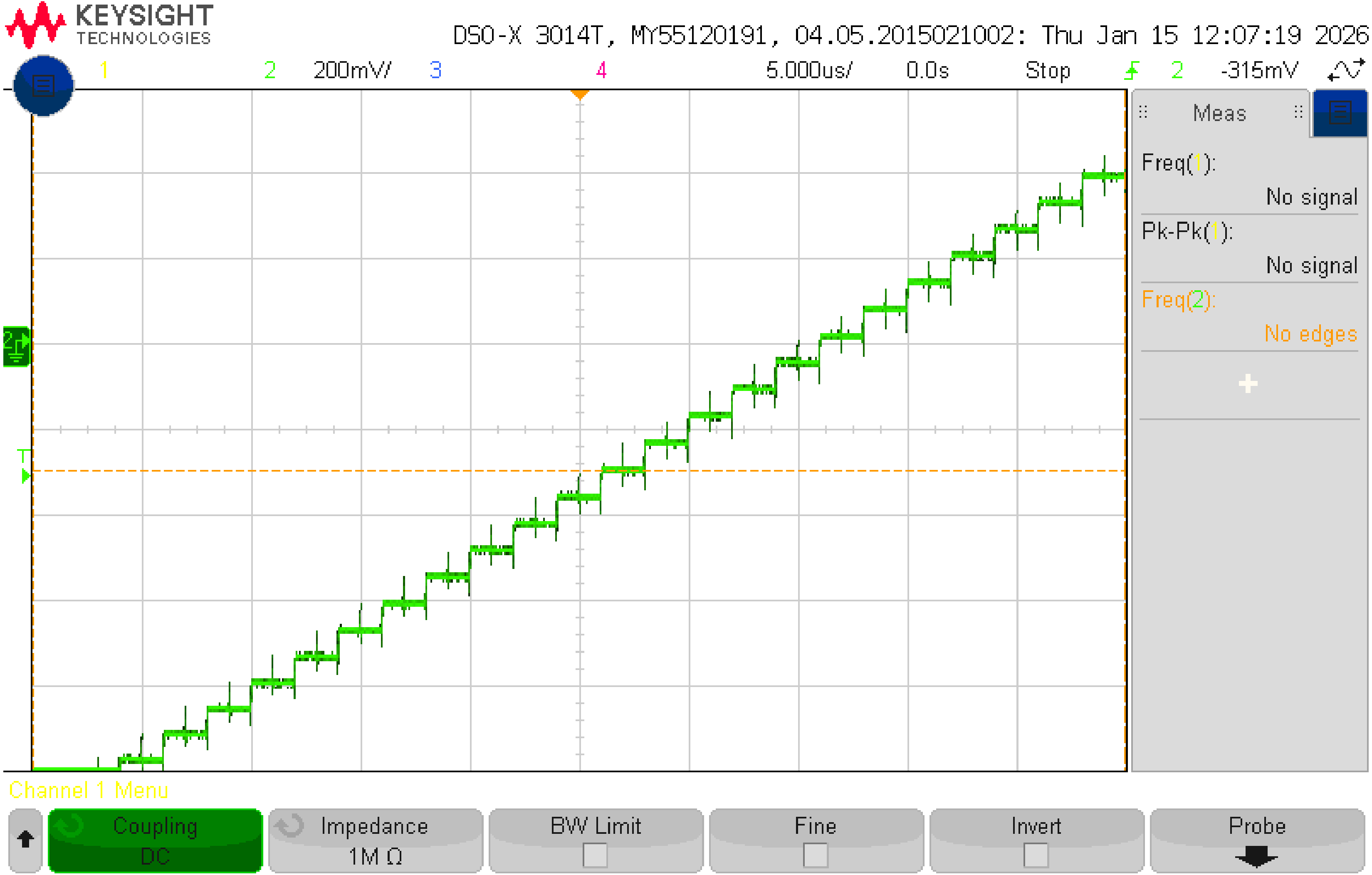


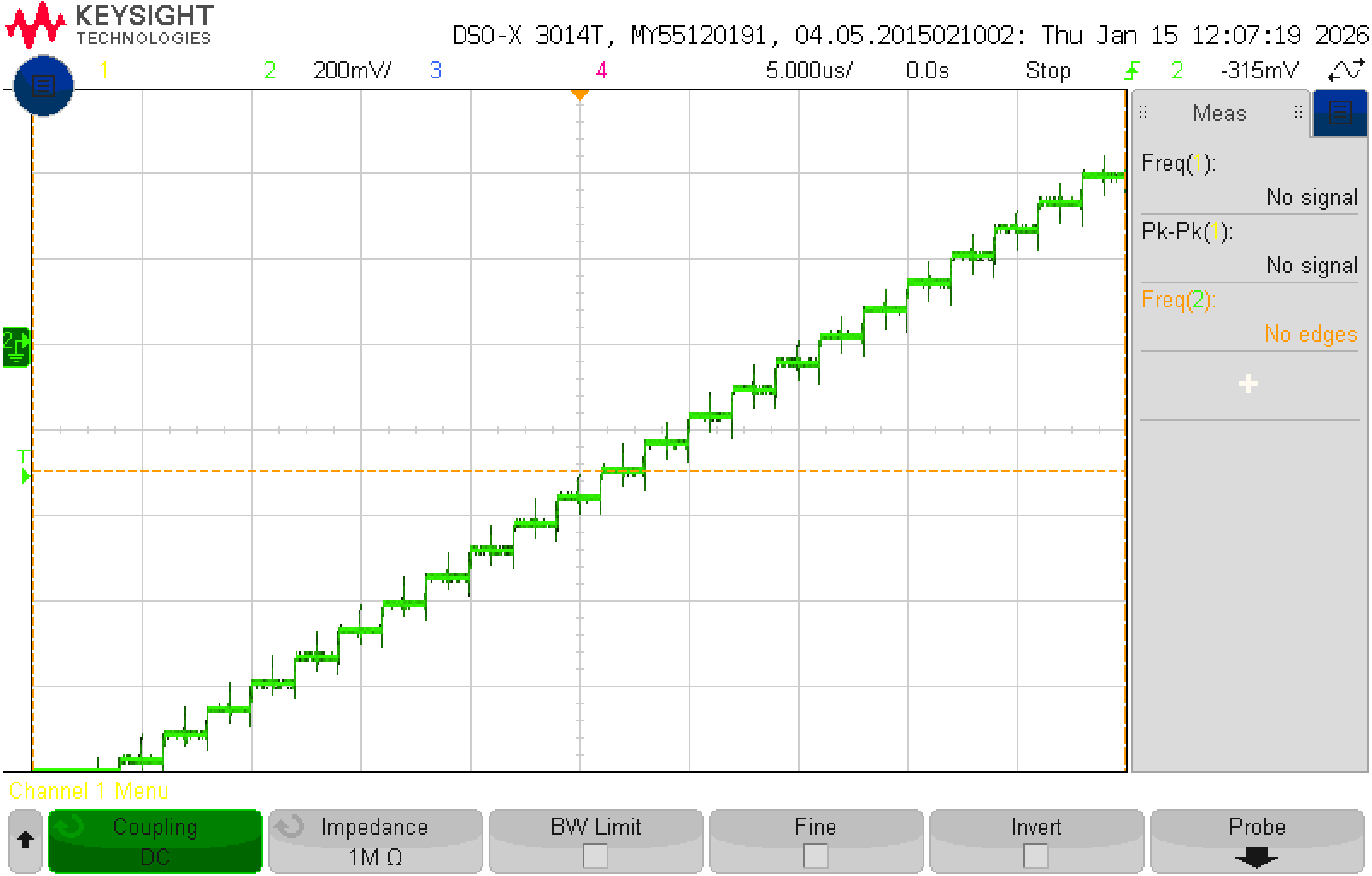
SC-filter out
C7
1863nF
R3
908
R4
9.04k
C6
4.7nF
0
R5
2693
C5
198pF
VCC
U2
LM4562
VINM
VINP
VCC
VEE
VOUT
1
2
3
4
5
0
C10
100n
VEE
Vout
R6
1k
R7
100k
VCC
C9
10uF
C8
100nF
0
U3
LM4562
VINM
VINP
VCC
VEE
VOUT
Vout2
VEE
VCC
V1
5Vdc
V5
5Vdc
C11
10u
VEE


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Thu Jan 22 11:20:21 2026
1
2 500mV/
3
4
100.0us/
0.0s
Auto
2 -244mV
Meas
Freq(2):
1.9960kHz
Ratio - Cyc(2→1):
No signal
Phase(2→1):
No signal
Phase(2→2):
0.0°
Autoscale Menu
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal

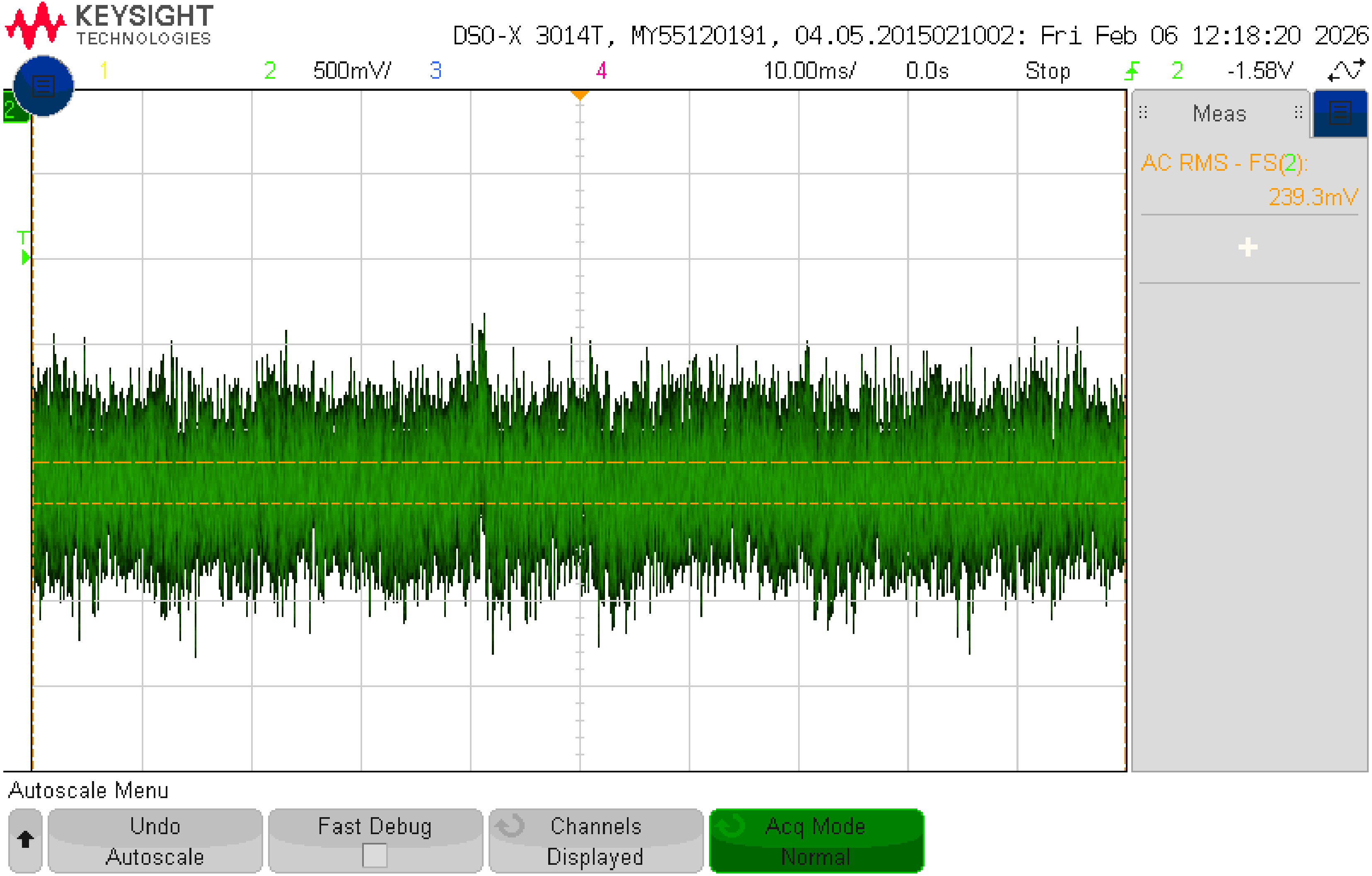

KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Thu Jan 15 10:45:13 2026
1 2.00V/ 2 3 4 500.0ns/ 0.0s Auto 1 2.44V
Meas
Freq(1):
500.00kHz
Pk-Pk(1):
5.8V
Measurement Menu
Meas Window: Auto Select
Source 1
Type: Freq
Add Measurement
Settings
Clear Meas
Statistics


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Thu Jan 15 11:58:59 2026
1 50mV/ 2 3 4 5.000us/ -38.00us Stop 1 -48.8mV
Meas
Freq(1):
1.146MHz
Pk-Pk(1):
161mV
-1.00mV DC 10.0 : 1
-2.50mV DC 10.0 : 1
0.0V DC 1.00 : 1
0.0V DC 1.00 : 1
11:58 AM
Jan 15, 2026


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Thu Jan 15 12:07:19 2026
1
2 200mV/
3
4
5.000us/
0.0s
Stop
2 -315mV
Meas
Freq(1):
No signal
Pk-Pk(1):
No signal
Freq(2):
No edges
Channel 1 Menu
Coupling
DC
Impedance
1M Ω
BW Limit
Fine
Invert
Probe



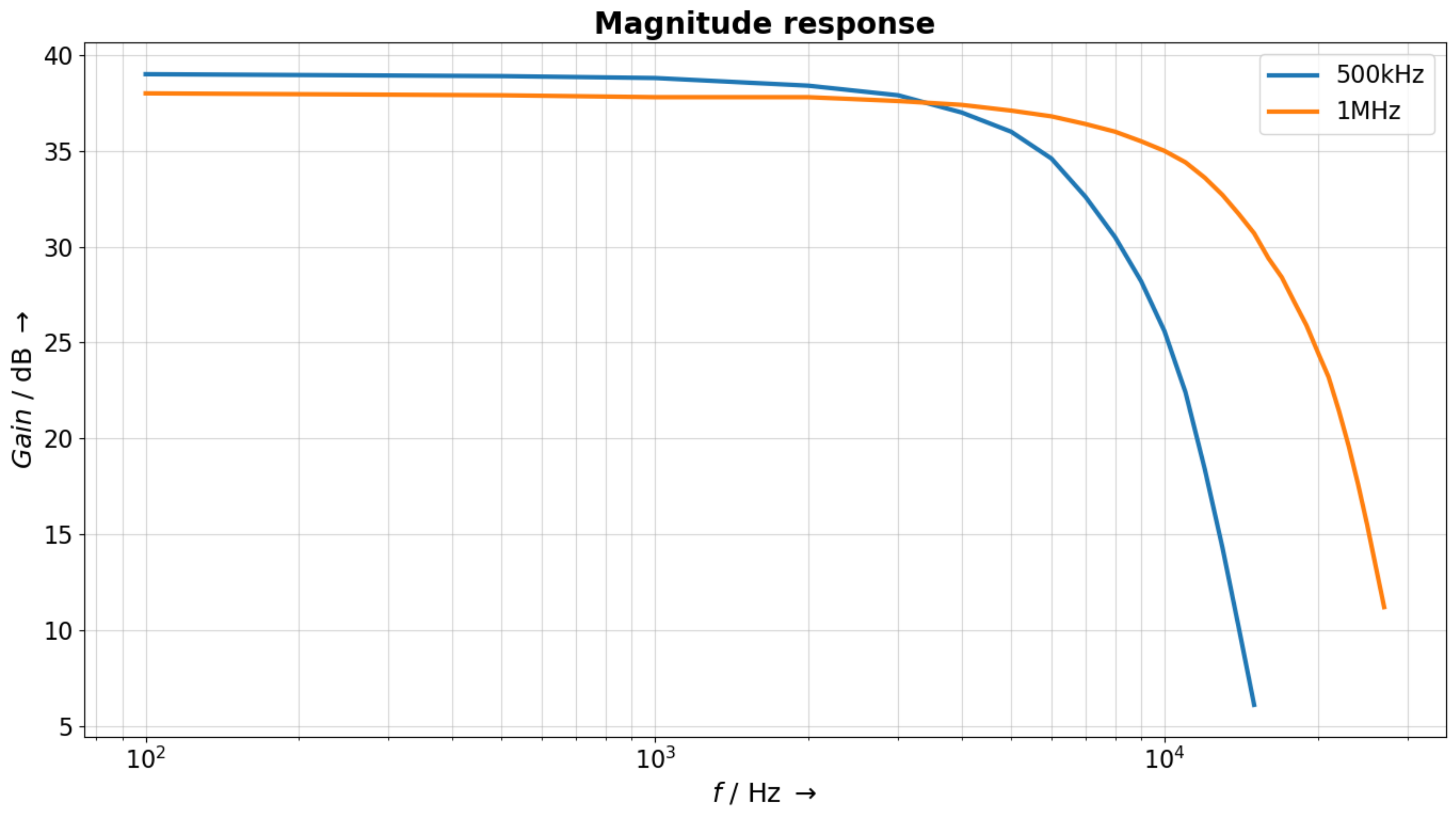
KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Thu Jan 15 12:07:19 2026
1
2 200mV/
3
4
5.000us/ 0.0s Stop 2 -315mV
Meas
Freq(1):
No signal
Pk-Pk(1):
No signal
Freq(2):
No edges
Channel 1 Menu
Coupling DC
Impedance 1M Ω
BW Limit
Fine
Invert
Probe


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Fri Feb 06 12:18:20 2026
1
2 500mV/
3
4
10.00ms/
0.0s
Stop
2 -1.58V
Meas
AC RMS - FS(2):
239.3mV
Autoscale Menu
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Fri Feb 06 12:14:19 2026
1
2 500mV/
3
4
2.000ms/
0.0s
Stop
2 -1.58V
Meas
AC RMS - FS(2):
216.2mV
Autoscale Menu
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Fri Feb 06 12:09:16 2026
1
2 200mV/
3
4
5.000ms/
0.0s
Stop
2 -1.40V
Meas
AC RMS - FS(2):
136.10mV
Autoscale Menu
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal


KEYSIGHT TECHNOLOGIES
DSO-X 3014T, MY55120191, 04.05.2015021002: Fri Feb 06 12:02:50 2026
1
2 200mV/
3
4
2.000ms/
0.0s
Stop
2 -1.47V
Meas
AC RMS - FS(2):
125.06mV
Autoscale Menu
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal



Magnitude response
500kHz
1MHz
Gain / dB →
40
35
30
25
20
15
10
5
10^2
10^3
10^4
f / Hz →

Table 4.1: Measurement results for the components

Component	**Selected**	**Measured**	**Calculated**
	R / kΩ, C / nF	R / kΩ, C / nF	R / kΩ, C / nF
R_G	0.05	0.051	/
R_1	1	0.9995	/
R_2	100	100.19	/
R_3	0.82	0.816	0.852
R_4	8.2	8.16	8.52
R_5	2.7	2.695	2.626
R_6	1	1.098	1
R_7	100	100.13	/
C_1	100	99.96	/
C_2	100	99.75	/
C_3	100	99.87	/
C_4	100	99.97	/
C_5	0.15	0.149	/
C_6	4.7	4.686	/
C_7	1500	1422.2	1868.1
C_8	100	99.97	/
C_9	10000	9821	/
C_{10}	10000	9851	/
C_{11}	100	99.94	/